

The Universal Language of Colors



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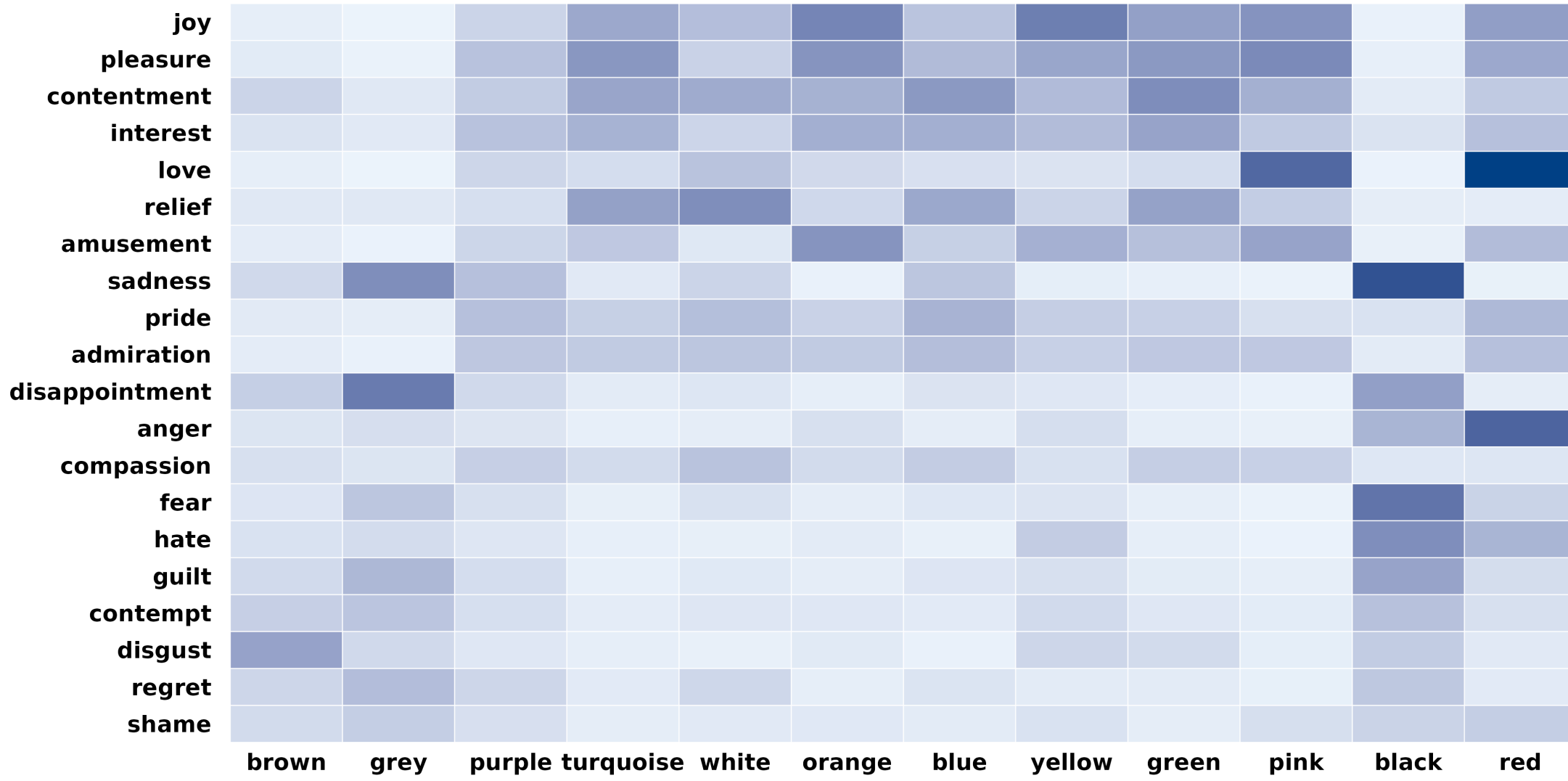
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The Psychology of Color: Universal or Cultural?

Colors are not merely visual stimuli; they are powerful emotional components that shape how we perceive the world around us. This study addresses a long-standing question in psychology: Are emotional responses to colors universal and biologically driven, or are they shaped by cultural context?

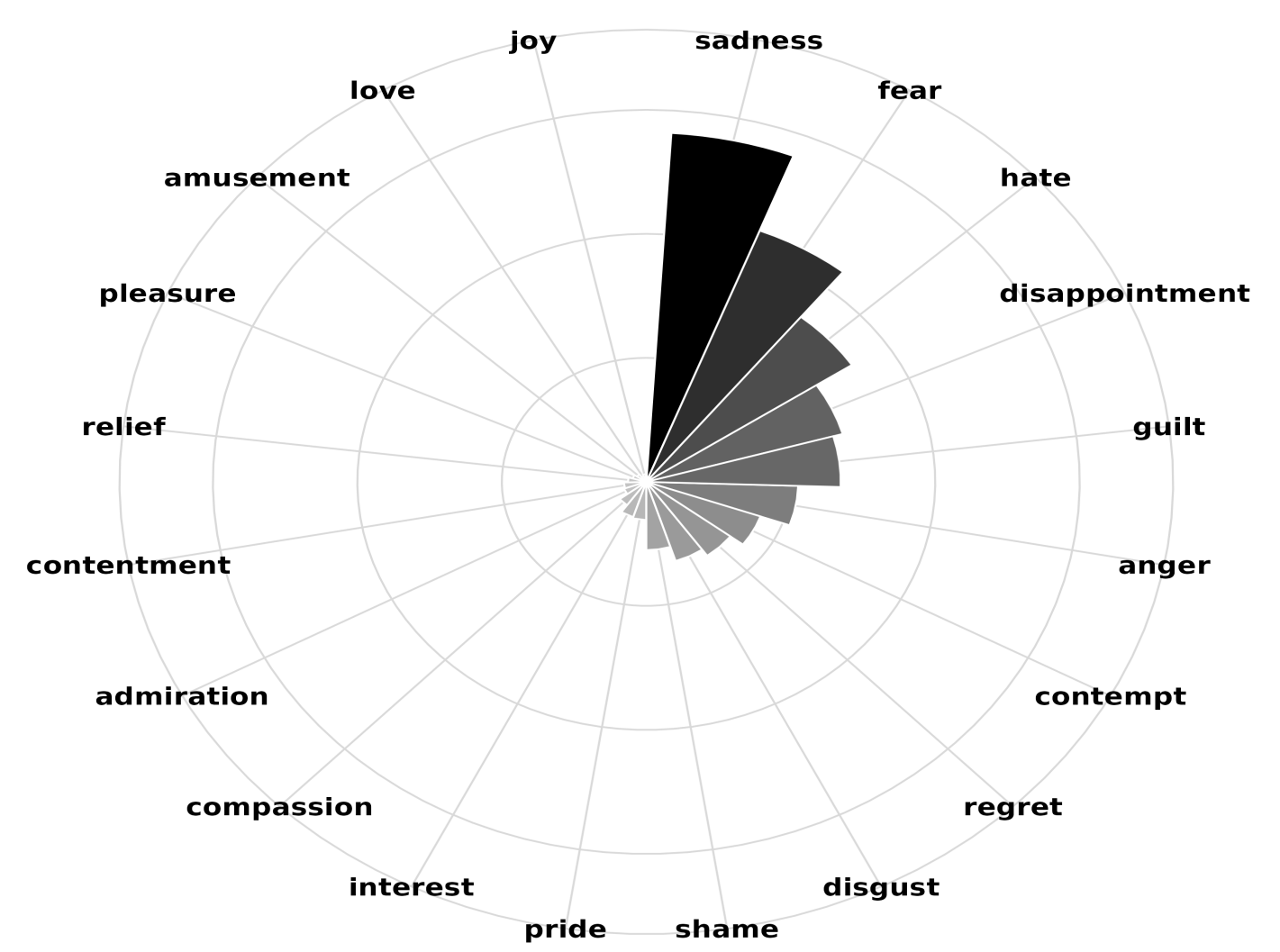
To explore this question, the project analyzes a reliable dataset published by Jonauskaitė et al. (2019) in *Royal Society Open Science*. The dataset includes 8,532 observations from 711 participants across four regions: China (representing Eastern culture) and Germany, Greece, and the United Kingdom (representing Western culture). By examining these cultural contrasts, this study aims to determine whether color–emotion associations reflect a universal language or a culture-specific experience.

Heatmap of Color-Emotion Associations



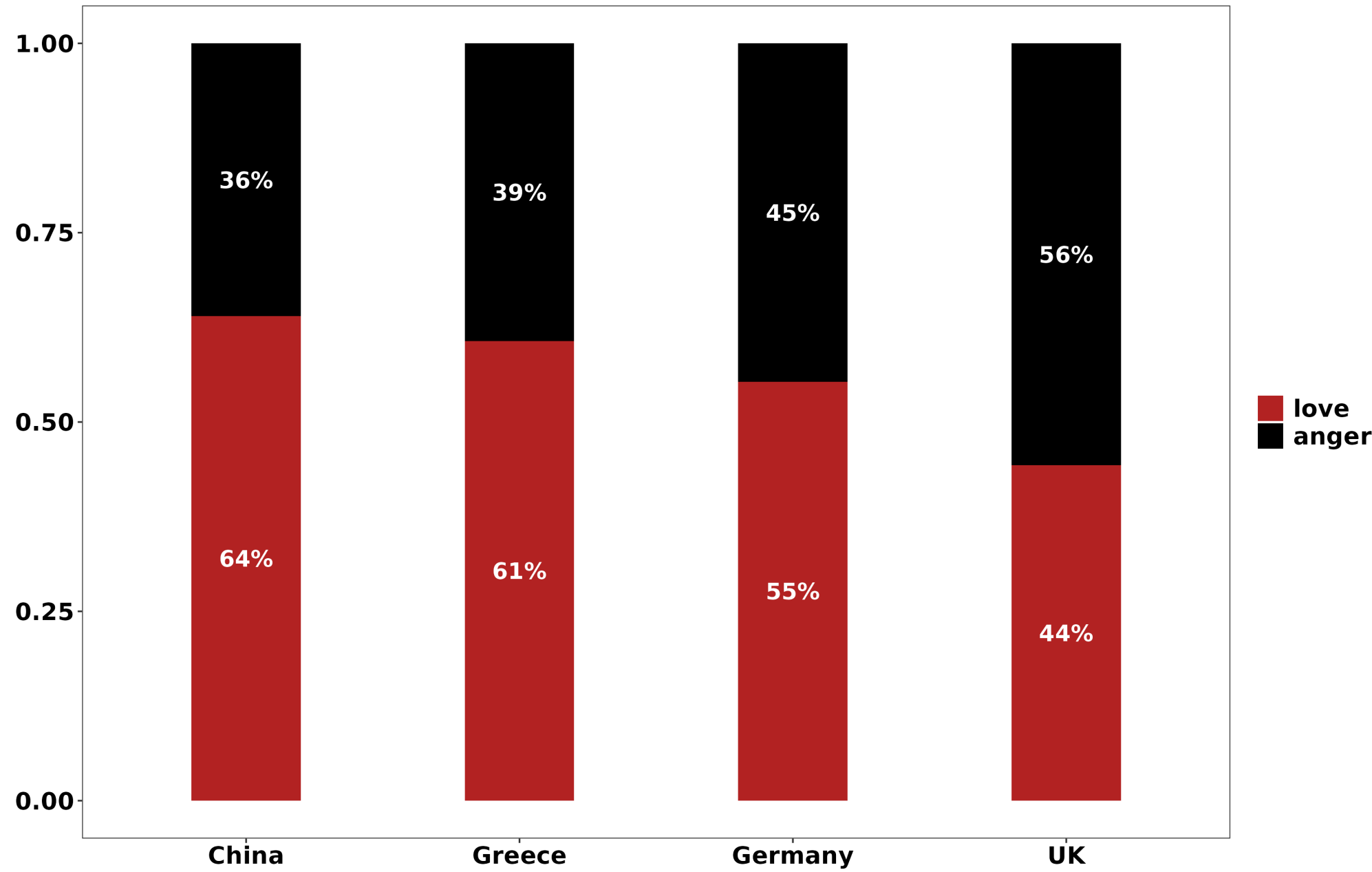
The heatmap reveals clear patterns in color–emotion associations, with negative emotions clustering around black, gray, and brown, and positive emotions around red, yellow, orange, and pink. Notably, red evokes both highly positive and highly negative emotions, while blue and green show more balanced emotional profiles.

The Emotional Depth of Black



The radar chart shows that the color black is strongly and consistently associated across cultures with negative emotions, particularly sadness, followed by fear, hate, and guilt. In contrast, black exhibits little to no association with positive emotions such as joy and love.

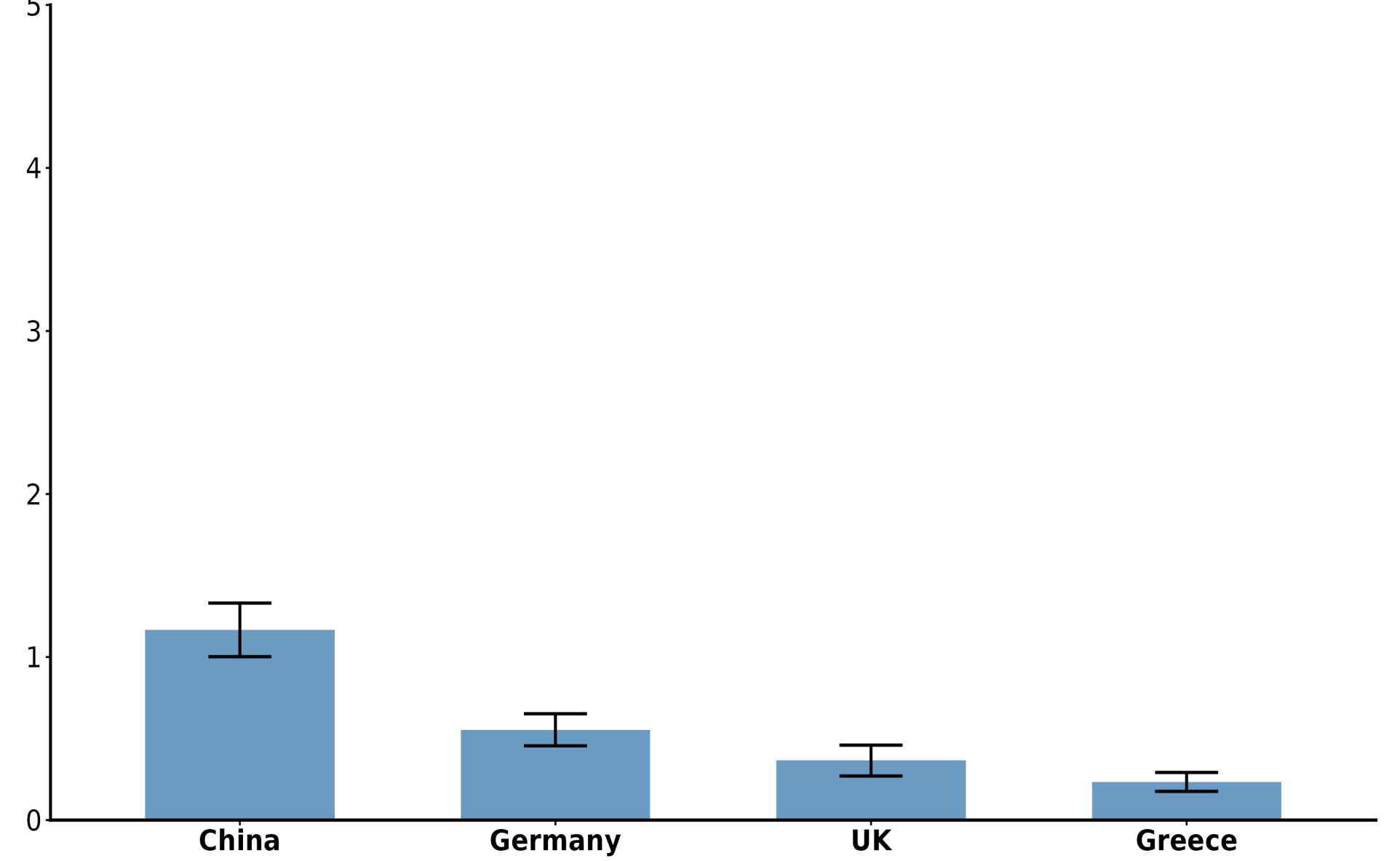
The Dual Nature of Red: Love and Anger



The bar chart highlights the dual emotional nature of the color red, which is associated with both love and anger across cultures. While red is predominantly linked to love in China, Greece, and Germany, the United Kingdom stands out as the only case where anger surpasses love. This pattern suggests that although the love–anger duality of red is widely shared, the dominant emotional response varies across cultural contexts.

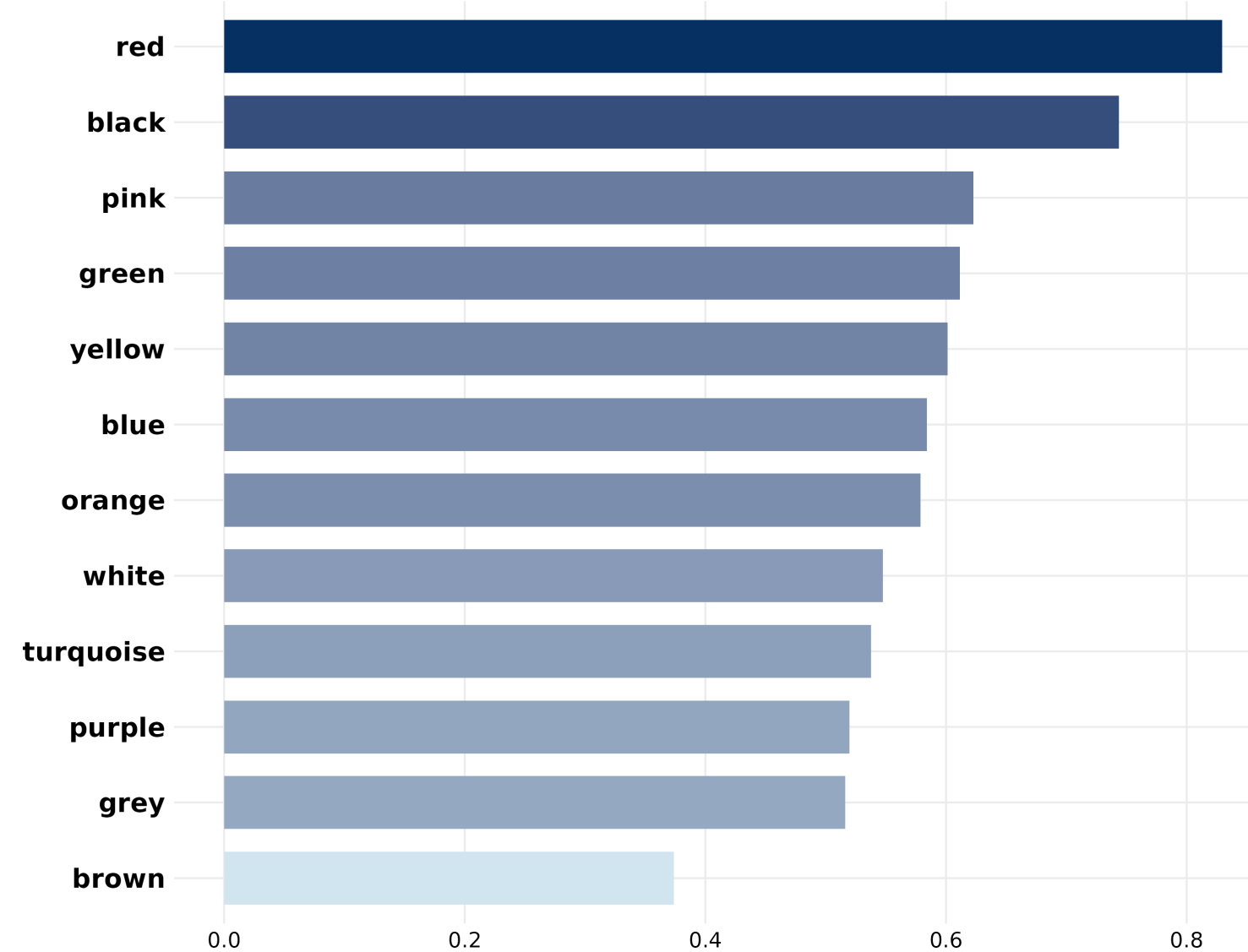
Sadness Intensity for White: Cultural Comparison

Error bars represent the Standard Error (SE)



The association between the color white and sadness varies markedly depending on cultural context. While this association is weak in Western countries, the strong linkage between white and sadness in China clearly highlights the decisive role of cultural background.

Global Emotional Intensity of Colors



Universal Hierarchy of Intensity: Red is the most emotionally charged color globally, evoking intense reactions in both positive (love) and negative (anger) spectrums. Black follows closely, driven by strong negative associations, while Brown and Grey are the least evocative. These rankings confirm a key insight: while the specific 'meaning' of a color may be cultural, its emotional 'power' is largely universal.

GENERAL CONCLUSION: UNIVERSAL INTENSITY VS. CULTURAL MEANING

Findings suggest that color–emotion associations are a combination of biological universality and cultural tradition.

1-Universal Intensity: Globally, Red and Black trigger the highest emotional impact, while Brown and Grey are the weakest, confirming a universal hierarchy of emotional intensity. Furthermore, Black is universally linked to sadness, implying a shared biological basis.

2-Cultural Context: Specific meanings are culture-bound. White signifies sadness in China, contrasting with Western neutrality. Similarly, the nature of red shifts: it is primarily love in China, but dominates as anger in the UK. **While emotional power is largely universal, specific emotional meaning is shaped by culture.**

